Hybrid Inference Network for Few-Shot SAR Automatic Target Recognition

Li Wang, Xueru Bai, *Member, IEEE*, Chen Gong, *Senior Member, IEEE*, and Feng Zhou, *Member, IEEE*

***Abstract*— Synthetic aperture radar (SAR) automatic target recognition (ATR) plays an important role in SAR image interpretation. However, at least hundreds of training samples are usually required for each target type in the existing SAR ATR algorithms. In this article, a novel few-shot learning framework named hybrid inference network (HIN) is proposed to tackle the problem of SAR target recognition with only a few training samples. The recognition procedure of HIN consists of two main stages. In the first stage, an embedding network is utilized to map the SAR images into an embedding space. In the second stage, a hybrid inference strategy that combines the inductive inference and the transductive inference is adopted to classify the samples in the embedding space. In the inductive inference section, each sample is recognized independently according to a metric based on Euclidean distance. In the transductive inference section, all samples are recognized as a whole according to their manifold structures by label propagation. Finally, in the hybrid inference section, the classification result is obtained by combining the above two inference methods. To train the framework more effectively, a novel loss function named enhanced hybrid loss is proposed to constrain samples to gain better interclass separability in the embedding space. Experimental results on the moving and stationary target acquisition and recognition (MSTAR) benchmark data set illustrate that HIN performs well in few-shot SAR image classification.**

***Index Terms*— Automatic target recognition (ATR), deep learning, few-shot learning (FSL), synthetic aperture radar (SAR).**

I. Introduction

THANKS to its all-day, all-weather, high-resolution, and long-distance operating capabilities, synthetic aperture radar (SAR) has been widely applied in battlefield reconnaissance, terrain mapping [1], geological exploration [2], and marine observation [3]. Particularly, SAR automatic target recognition (ATR) has attracted much attention due to its military significance. However, unlike optical imaging, single-polarized SAR image represents the scattering intensity of the target by gray scales and usually has blurred edges and strong anisotropy due to limited resolution and background clutters [4]. All these factors increase the difficulty of effective feature extraction and target recognition.

There are three mainstream methods, i.e., template matching [5], [6], target modeling [7], [8], and machine learning [9]–[13]. Such methods require the designing of a dedicated template, a target model, or a classifier in advance. Nevertheless, heavy reliance on hand-designed features usually results in high complexity and poor generalization performance. Deep learning model is one of the data-driven machine learning methods that can automatically extract features from samples. In recent studies, it is found to be highly effective in SAR ATR tasks [14]–[22]. It should be noted that these methods need at least hundreds of training samples for each target type, and their recognition performance will degrade significantly if there are only a few training samples in some classes. Lin *et al.* [22] demonstrated that the recognition accuracy of deep learning methods and traditional machine learning methods will fall below 40% when the training set only includes dozens of samples for each class (about 10% of the moving and stationary target acquisition and recognition (MSTAR) standard operating conditions (SOC) training set). In the real-world situation, it may be difficult and costly to obtain hundreds of SAR images for some types of targets. More extremely, under certain circumstances, e.g., military reconnaissance, only a few SAR images of some special targets can be obtained and the existing SAR ATR algorithms are not effective anymore.

The starvation of labeled samples challenges the existing SAR ATR methods that lack the ability of learning from very few labeled samples and fast generalizing to new tasks. Inspired by humans' ability to recognize with only a few samples and motivated by the prohibitive cost of the collection of training samples, the research area of few-shot learning (FSL) has received increasing interests [23]. Different from the traditional classification methods, in which hundreds of labeled samples or more are needed for each target class to be recognized, FSL aims to perform classification under the restriction that only a few labeled examples of each possible class can be observed before making a prediction about a test instance. However, there is no research on SAR ATR based on FSL so far.

In FSL, three data sets are considered: the query set, the support set, and the training set. The query set contains the unlabeled samples of the classes that need to be recognized. The support set contains the same classes of targets as the query set, with only a few labeled samples in each class.

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Li Wang and Feng Zhou are with the Ministry Key Laboratory of Electronic Information Countermeasure and Simulation, Xidian University, Xi'an 710071, China (e-mail: fzhou@mail.xidian.edu.cn).

Xueru Bai is with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China (e-mail: xrbai@xidian.edu.cn).

Chen Gong is with the Department of Electronic and Information Science, University of Science and Technology of China, Hefei 230026, China.

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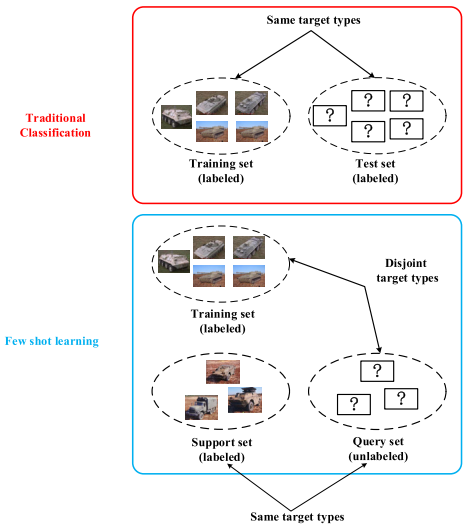


Fig. 1. Data sets in traditional classification and FSL.

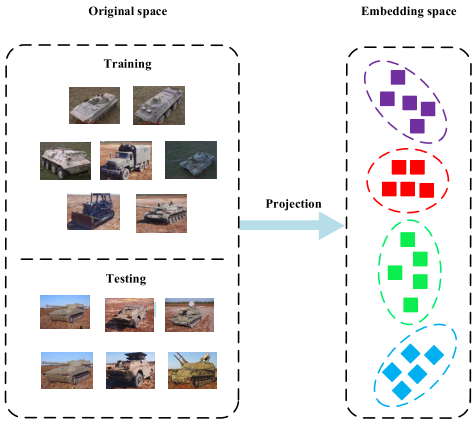


Fig. 2. Intuitive example of the projection in FSL.

The training set contains classes disjoint from the support set. A comparison between the data sets utilized in a traditional classification task and an FSL task is shown in Fig. 1.

Most existing FSL approaches exploit a representation shared between the auxiliary training set and the support set. In order to get the representation, a projection from the original sample space to a new embedding space where the classification problem becomes easier is learned from the training set. Fig. 2 gives an intuitive illustration of the projection. Recently, FSL methods using deep neural networks to learn the projection have achieved great success thanks to the superiority of automatic feature extraction and powerful representation [24]–[27].

After the projection is learned, it is used in the test stage to map each sample into the same embedding space where the classification is performed. Most existing FSL methods adopt inductive inference at this stage, i.e., predicting query samples one by one according to certain rules that define the similarity between samples in the query set and the support set. However, these approaches rely heavily on the representativeness of the samples in the support set. Different from the traditional classification problem where large amounts of labeled training samples are utilized to construct a robust classification model, any sample in the support set that does not represent its corresponding category well will decrease the credibility of classification in FSL. To solve this problem, transductive propagation network (TPN) is proposed [28]. It adopts transductive inference on the classification stage by label propagation that considers relationships between samples in the query set and predicts them as a whole. However, the classification performance of TPN relies heavily on the manifold structure of unlabeled samples in the embedding space because the key of label propagation is to classify the close samples in the embedding space into the same category. In practice, the manifold structure of unlabeled samples that is convenient for classification does not always exist. Therefore, the high credibility of transductive inference cannot be guaranteed.

In order to overcome the shortcomings discussed above, we propose a hybrid inference strategy that not only considers the similarity between the samples in the support set and the query set but also exploits the manifold structure of the query set in the embedding space. According to the classification result of the hybrid inference, a hybrid loss mathematically similar to the cross-entropy loss is proposed. However, it is found that if the hybrid loss is directly utilized to train the framework, a high training accuracy in the learned embedding space cannot always be obtained from inductive inference or transductive inference. Such phenomenon indicates that the hybrid loss reduces the difficulty of recognition, but a good embedding space where the samples can be well classified through both inductive inference and transductive inference cannot be learned. If the inference result based on inductive inference or transductive inference is not credible, the hybrid inference result based on them is not credible either. To solve this problem, an enhanced hybrid loss with two extra constraint items is proposed.

In order to learn large amounts of unknown parameters, most existing FSL algorithms based on deep learning adopt the episode-based training method that requires the training set to contain sufficient target classes to ensure good generalization performance [25]. However, it is difficult to obtain a large number of SAR images and the number of training samples in a few-shot SAR ATR task is much less than that of a typical FSL task. Thus, the typical episode-based training method is not effective anymore. To overcome this problem, we propose a random episode-sampled method to ensure the effectiveness of training.

The main contribution of this article can be summarized as follows.

1) A novel FSL method called hybrid inference network, i.e., HIN is proposed. By adopting a hybrid inference strategy, HIN overcomes the deficiency of singly adopting inductive inference or transductive inference in FSL and improves the recognition performance.
2) An enhanced hybrid loss is proposed to constrain the embedding network to learn a good projection that can map samples to an embedding space where both inductive inference and transductive inference perform well.

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3) To alleviate the starvation of training samples in few-shot SAR ATR, a random episode-sampled method is proposed to train the learnable parameters in HIN, which requires a much smaller training set than the existing episode-based training method.
4) For three-class SAR ATR problems, the recognition accuracy of HIN is similar to those of traditional SAR ATR methods, but HIN requires many fewer labeled samples than traditional SAR ATR methods.

II. Related Work

A. FSL Based on Embedding Learning

The mainstream FSL methods adopt an embedding learning framework that consists of two stages. In the first stage, a projection function that maps samples from their original space to a new embedding space is learned. In recent studies, many methods take deep neural network as the projection function due to its strong ability of representation and automatic feature extraction. In the second stage, a classifier is utilized to classify samples in the new feature space. Various metric criteria are utilized to construct classifiers, such as squared l_2 distance [25], [29], [30], cosine similarity [24], [31], weighted l_1 distance [32], and Gaussian similarity [28].

B. Inductive Inference and Transductive Inference

Inductive inference and transductive inference are two typical inference methods in machine learning theory. For inductive inference, a certain rule is obtained by analyzing the labeled training samples. Then, each unlabeled sample is predicted independently according to the rule (e.g., nearest neighbor classifier). For transductive inference, the unlabeled samples are predicted as a whole (e.g., label propagation [35]). By exploiting the manifold structure of the unlabeled samples, the generalization performance of the model is improved. When there are few labeled training samples and many unlabeled samples, the recognition performance of inductive inference is usually poor. Transductive inference can utilize the manifold information of unlabeled samples and perform more reliable classification. However, transductive inference relies heavily on the manifold structure of unlabeled samples. If the unlabeled samples do not have a good manifold structure (e.g., there are too few unlabeled samples), it is difficult to obtain good classification performance by transductive inference. The comparison between inductive inference and transductive inference is shown in Fig. 3. The colored blocks represent labeled samples and the white blocks represent unlabeled samples. The solid lines between samples indicate that the classification result depends on the relationship between them. For inductive inference in Fig. 3(a), each unlabeled example is predicted independently according to labeled samples. For transductive inference in Fig. 3(b), each unlabeled example is predicted according to both labeled samples and other unlabeled samples.

III. Hybrid Inference Network

This section introduces the proposed FSL method named HIN that consists of three main stages, as shown in Fig. 4.

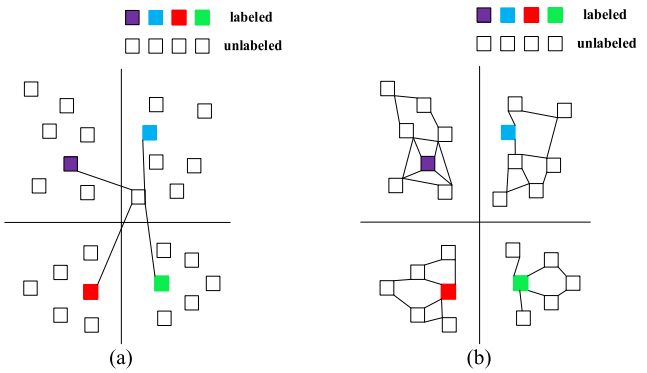


Fig. 3. Inductive inference versus transductive inference. (Left) For inductive inference, each unlabeled example is predicted independently according to labeled samples. (Right) For transductive inference, the unlabeled samples are predicted entirely. (a) Inductive inference. (b) Transductive inference.

In the training data generating stage, in order to reduce the amount of data the typical episode-based FSL training method requires, a random generating algorithm is designed to generate sample/query set from the training set. In the training stage, first, an embedding network is utilized to map the original SAR images to a new embedding space. Second, three different inference methods, i.e., inductive inference, transductive inference, and hybrid inference, are adopted to predict unlabeled samples, respectively. Third, their corresponding cross-entropy loss functions are calculated with the help of the true label. Finally, by summing over the above three cross-entropy loss functions, an enhanced hybrid loss is obtained to update the parameters of HIN end-to-end. In the test stage, all parameters of HIN are fixed and the classification result is output by hybrid inference. The details of HIN are described as follows.

A. Problem Setup and Training Data Generating

In a typical FSL task, there are three original data sets, i.e., the query set, the support set, and the training set. The query set contains the classes of targets that need to be recognized. The support set contains the same classes of targets as the query set, with only a few labeled samples in each class. It is worth noting that the original support set and test set are only utilized in the test stage, and the training set contains classes of targets disjoint from the original support set. If the support set contains K labeled samples for each of C unique classes, the corresponding FSL problem is called a C-way K-shot FSL problem. Due to the lack of labeled samples in the support set, a classifier trained by only the support set does not have satisfying performance. FSL aims to extract transferrable knowledge that will help the FSL algorithm to perform better from an auxiliary training set and classify the unlabeled query samples more correctly.

Episode-based training method is an effective way to exploit the training set to obtain effective transferrable knowledge [25]. The key of episode-based training method is to mimic test tasks by sampling training tasks (episodes). For a C-way K-shot FSL problem, an episode is obtained by randomly selecting C classes from the training set in each training

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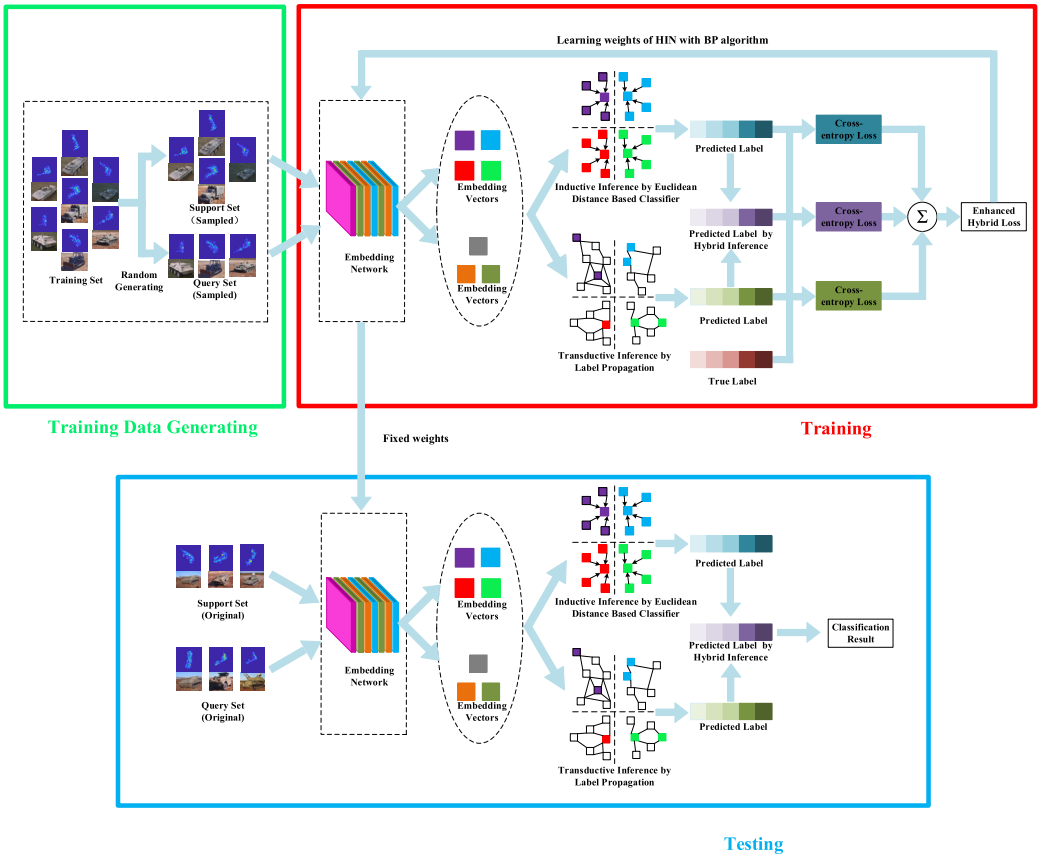


Fig. 4. Framework of HIN.

iteration. Then, K labeled samples are randomly selected from each of the C classes to constitute the sampled support set, while the remaining samples of the C classes constitute the sampled query set. The splitting of the sampled support/query set aims to simulate the original support/query set that will be encountered in the test stage. To avoid confusion, the support/query set generated from the training set will be annotated by "sampled" support/query set, while the support/query set without annotations refers to the original support/query set.

The typical episode-based training method requires that the training set should contain sufficient target classes to ensure good generalization performance. Omniglot and miniImageNet are two benchmarks for evaluating FSL algorithms, which contain 1623 and 100 classes of targets, respectively (where 1200 and 64 classes of targets serve as training sets, respectively). However, it is difficult to obtain so many classes of SAR images in a few-shot SAR ATR task as in a typical FSL task. Thus, the typical episode-based training method is not effective any more. To solve this problem, a random-episode data generating method is proposed, as shown in Fig. 5, to ensure the effectiveness of the framework training. For a C-way K-shot SAR ATR task with a training set containing M categories and N samples for each category, the random-episode data generating method can be described as follows. In each training episode, C categories of samples are randomly selected from the training set. Then, K samples are selected from each of these C categories to serve as the sampled support set, while other samples in the selected C categories serve as the sampled query set. Theoretically, a total of

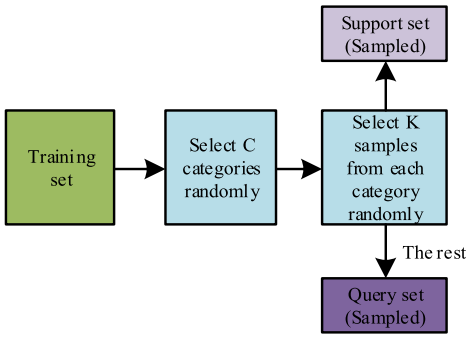


Fig. 5. Random sampled support/query set generation in a training episode.

$[M!/C!(M-C)!] \cdot [N!/K!(N-K)!]^C$ different sampled support/query sets will be generated from the above method. Such a large number can ensure the diversity of the sampled support/query set for a small M. Thus, the generalization performance of the model is improved. In each episode during training, Adam algorithm [33] is applied to update the weights once and the batch size is the size of the full query set. Such procedure is repeated until the network converges.

B. Embedding Network

As shown in Fig. 4, the embedding network plays an important role in the proposed method by mapping each SAR image from the original space to a new space through the embedding function $f_\phi : \mathbb{R}^D \rightarrow \mathbb{R}^L$, where ϕ denotes the learnable

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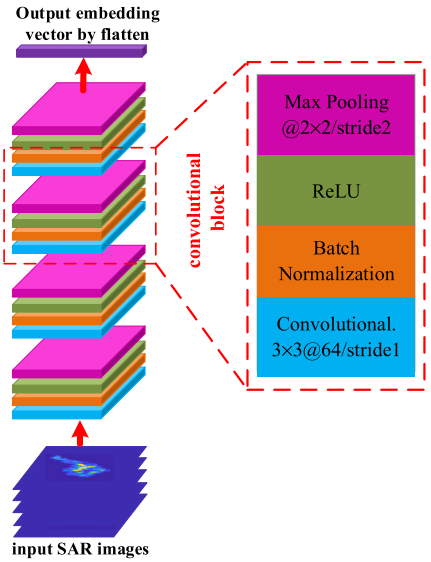


Fig. 6. Structure of the embedding network.

parameters. The embedding network reduces the difficulty of classification by making similar samples close together while separate dissimilar samples. Usually, L is smaller than D. In HIN, a deep convolutional network is utilized as the embedding network owing to its strong automatic feature extraction capability.

Fig. 6 shows the structure of the embedding network that consists of four convolutional blocks. Each convolutional block consists of a convolutional layer, a rectified linear unit (ReLU) nonlinearity layer, a batch normalization layer, and a max pooling layer. The convolutional layer contains 64 convolutional filters and they share the same size of 3×3. The ReLU nonlinearity layer and the batch normalization layer are added after the convolutional layer, respectively. The max pooling layer with the kernel size of 2×2 and the stride size of 2 is added after the batch normalization layer. Finally, the embedding vector is obtained by flattening the feature maps of the last convolutional block. In Section IV-E, visualization of the SAR samples in the original space and embedding space is given.

C. Inductive Inference by the Euclidean Distance-Based Classifier

As shown in Fig. 3, when inductive inference is adopted, each unlabeled example is predicted independently according to certain metrics, and such operation has nothing to do with other unlabeled samples. Because the Euclidean distance is a classical and effective metric [25], [34] in FSL tasks, the proposed method adopts Euclidean distance-based classifier for inductive inference. It should be noted that the design of the two subclassifiers, which are based on inductive inference and transductive inference, respectively, needs to be considered as a whole. The two subclassifiers should promote each other and improve the accuracy rate together. The label propagation-based classifier also benefits from the constraint

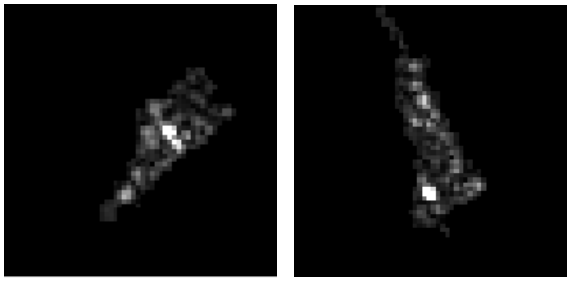


Fig. 7. Two SAR images of BMP-2 infantry fighting vehicle.

of the Euclidean distance because it utilizes the Euclidean distance to construct the manifold structure.

The details of the Euclidean distance-based classifier are described as below. For each episode in a C-way K-shot FSL problem, the support set $S = \{(x_1, y_1), \ldots, (x_N, y_N)\}$ with $N = C \times K$ labeled samples is given and $y_i \in \{1, ..., K\}$ is the corresponding label. S_k denotes the set of samples labeled with class k in the support set. Each prototype $c_k \in R^L (k \in 1, 2, \ldots, C)$ is obtained by calculating the clustering center of each class of the support set in the embedding space as follows:

$$c_k = \frac{1}{|S_k|} \sum_{(x_i, y_i) \in S_k} f_\phi(x_i). \tag{1}$$

Then, the Euclidian distance between the embedding vector of each sample in the query set and each prototype is calculated. Finally, inductive inference is conducted by the Euclidean distance-based classifier

$$p_E(y_i = k|x_i) = \frac{\exp\left(-d\left(f_\phi(x_i), c_k\right)\right)}{\sum_k^C \exp\left(-d\left(f_\phi(x_i), c_k\right)\right)} \tag{2}$$

where $p_E(y_i = k|x_i)$ is the probability of a sample x_i being classified to the kth category, $f_\phi(x_i)$ is the embedding vector output by the embedding network, c_k is the prototype of the kth category calculated by (1), and $d(f_\phi(x_i), c_k)$ is the Euclidean distance between $f_\phi(x_i)$ and c_k, i.e., $d(f_\phi(x_i), c_k) = ||f_\phi(x_i) - c_k||^2$.

Because the mathematical expression of the Euclidean distance-based classifier is similar to that of the softmax classifier, for query samples $\{(x_i, y_i), i = 1, \ldots, m\}$ in each training episode, the corresponding cross entropy loss can be expressed as follows:

$$L_E = -\frac{1}{m} \sum_i \log p_E(y = y_i|x_i). \tag{3}$$

In classification tasks, we can expect that the two samples belong to the same class if their Euclidean distance is small. However, it is not a necessity that the Euclidean distance is small for the two samples in the same class. As an illustration, the images in Fig. 7 have a large Euclidean distance although they belong to the same class. In a few-shot classification problem, if the diversity of samples in the support set is limited, then the recognition performance of the Euclidean distance-based classifier is not satisfying. To tackle this problem, the similarity of two samples in the proposed method is

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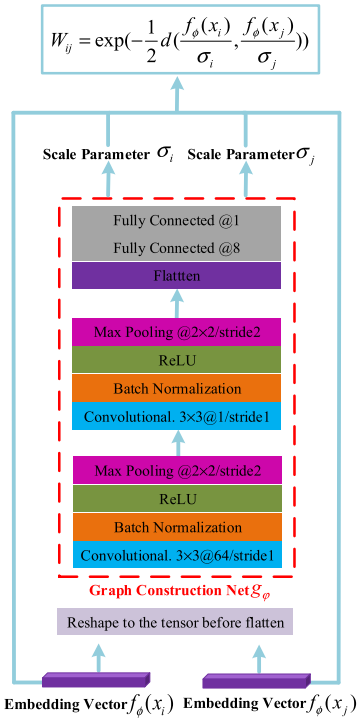


Fig. 8. Computing scale parameter with the graph construction net.

measured by the distance of their embedding vectors, and the Euclidean distance-based loss L_E constrains the samples of the same class to get close in the embedding space.

D. Transductive Inference Based on Label Propagation

In order to overcome the shortcoming of inductive inference, transductive inference that conducts inference in the whole query set is proposed. Label propagation is an effective transductive inference method that exploits the manifold structure of a data set by constructing graph weights between samples with Gaussian similarity

$$W_{ij}=\exp\left(-\frac{d(x_i,x_j)}{2\sigma^2}\right) \tag{4}$$

where $d(\cdot)$ is a distance metric (e.g., Euclidean distance) and σ is the scale parameter. Various parameters σ lead to quite different graph structures. Usually, it is difficult to choose the optimal σ in label propagation [36]. In this article, a graph construction net g_φ with learnable parameters φ is used to compute σ and the framework is shown in Fig. 8. In order to make full use of the support set with limited labeled samples and conduct transductive inference in the query set, a graph is built on the union set of support set and query set. For any two samples x_i and x_j in the union set, their embedding vectors are represented by $f_\phi(x_i)$ and $f_\phi(x_j)$. First, $f_\phi(x_i)$ and $f_\phi(x_j)$ are encoded into two scale parameters by a graph construction net which consists of two convolutional blocks (64 filters and 1 filter) and two fully connected layers (8 nodes and 1 node). When the embedding vector $f_\phi(x_i)$ is input into the graph construction net, it is reshaped to a tensor which shares the same size with the tensor before flattening in the embedding network. Then, the graph weights are computed by

$$W_{ij}=\exp\left(-\frac{1}{2}d\left(\frac{f_\phi(x_i)}{\sigma_i},\frac{f_\phi(x_j)}{\sigma_j}\right)\right). \tag{5}$$

For a C-way-K-shot few-shot classification problem with T samples in the query set, we need to calculate the graph weight between any two samples, i.e., $W\in R^{(C\times K+T)\times(C\times K+T)}$ for all the samples in the union set. However, there are usually a large number of unlabeled samples in the query set and W becomes very large. Based on the assumption that each sample is only related to the samples close to it in the embedding space, a k-nearest neighbor graph is constructed by only keeping the k max values in each row of W. Then, the normalized graph Laplacian of W can be calculated by

$$S=D^{-1/2}WD^{1/2} \tag{6}$$

where D is a diagonal matrix with D_{ii} being the sum of the ith row of W. Once the graph is constructed, transductive inference can be conducted via label propagation.

Matrix $F\in R^{(C\times K+T)\times C}$ is defined and each row of F denotes the label of each sample in the union set. In the beginning, another label matrix $Y\in R^{(C\times K+T)\times C}$ is defined with $Y_{ij}=1$ if x_i is from the support set and labeled as $y_i=j$ and $Y_{ij}=0$ if otherwise. Then, F can be iteratively updated using the following formulation:

$$F_{t+1}=\lambda SF_t+(1-\lambda)Y \tag{7}$$

where F_t denotes the predicted result at time step t, S is the normalized graph Laplacian, and $\lambda\in(0,1)$ is a hyperparameter that controls the amount of information updated in each time step. Zhou *et al.* [35] proved that sequence $\{F_t\}$ converges to

$$F^*=(I-\lambda S)^{-1}Y \tag{8}$$

where I is the identity matrix. In order to make the inference procedure more efficient, this result is directly used in HIN. Finally, transductive inference is conducted by label propagation

$$p_{lp}(y_i=k|x_i)=\frac{\exp(F^*_{ik})}{\sum_{k=1}^{C}\exp(F^*_{ik})} \tag{9}$$

where $p_{lp}(y_i=k|x_i)$ is the probability of the sample x_i being classified to the kth category predicted by label propagation.

Similar to the inductive inference, for query samples $\{(x_i,y_i),i=1,\ldots,m\}$ in each training episode, the corresponding cross entropy loss of label propagation can be expressed as

$$L_{lp}=-\frac{1}{m}\sum_i\log p_{lp}(y=y_i|x_i). \tag{10}$$

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E. Hybrid Inference

As discussed above, the classification performance of Euclidean distance-based classifier and label propagation relies heavily on the representativeness of samples in the support set and the manifold structures of the query set, respectively. Motivated by seeking a new inference strategy that can overcome these disadvantages, the following hybrid inference method is proposed:

$$\begin{aligned} p_h(y_i = k|x_i) &= \alpha p_E(y_i = k|x_i) + \beta p_{lp}(y_i = k|x_i) \\ &= \frac{\alpha \exp\left(-d\left(f_\phi(x_i), c_k\right)\right)}{\sum_{k=1}^{C} \exp\left(-d\left(f_\phi(x_i), c_k\right)\right)} + \frac{\beta \exp\left(F_{ik}^*\right)}{\sum_{k=1}^{C} \exp\left(F_{ik}^*\right)} \end{aligned} \tag{11}$$

where p_E and p_{lp} are the classification results of (2) and (9); α and β are the parameters that adjust the mix ratio of inductive inference and transductive inference. For an arbitrary sample in a C-class classification problem, (2) and (9) will output C soft labels, respectively, i.e., $\{a_1, a_2, \ldots, a_C\}$ and $\{b_1, b_2, \ldots, b_C\}$. The following inference strategy is utilized to compute α and β:

$$\alpha = \frac{\max\{a_1, a_2, \ldots, a_C\}}{\max\{a_1, a_2, \ldots, a_C\} + \max\{b_1, b_2, \ldots, b_C\}} \tag{12}$$

$$\beta = \frac{\max\{b_1, b_2, \ldots, b_C\}}{\max\{a_1, a_2, \ldots, a_C\} + \max\{b_1, b_2, \ldots, b_C\}} \tag{13}$$

where $\max\{a_1, a_2, \ldots, a_C\}$ and $\max\{b_1, b_2, \ldots, b_C\}$ represent the confidences of the two inference methods in their classification results, respectively.

As a matter of course, for query samples $\{(x_i, y_i), i = 1, \ldots, m\}$ in each training episode, the corresponding cross entropy loss of hybrid inference can be expressed as

$$L_h = -\frac{1}{m} \sum_i \log p_h(y = y_i|x_i). \tag{14}$$

F. Enhanced Hybrid Loss

In the experiments, we found that if (14) is directly used as the objective function to train HIN, (11) will have a high training accuracy, while (2) and (9) often lead to lower training accuracies, respectively. Such phenomenon indicates that (11) reduces the difficulty of classification but cannot constrain the embedding network to map SAR images to a new embedding space where both Euclidean distance-based classifier and label propagation perform well. To solve this problem, an enhanced hybrid loss is proposed as the final objective function

$$L_{eh} = L_E + L_{lp} + L_h. \tag{15}$$

By adding L_E and L_{lp} directly to the objective function, the classification performance of the Euclidean distance-based classifier and label propagation will be improved in the embedding space. The effectiveness of the enhanced hybrid loss will be demonstrated in the following experiments.

TABLE I
DEPRESSION ANGLE SETTINGS

	Depression		
	Training set	Support set	Query set
Experiment 1	17°	17°	15°
Experiment 2	17°	17°	30°

IV. EXPERIMENT

A. Data Set

The training set, support set, and query set utilized in this article are generated from the MSTAR [37] data set which is a benchmark data set for SAR ATR research provided by DARPA/AFRL. The data set was collected by Sandia National Laboratory SAR sensor platform in 1995 and 1996 using an X-band SAR sensor. It provides a nominal spatial resolution of 0.3 m × 0.3 m in both range and azimuth directions, and the image size is 128 × 128. The publicly released data sets include ten categories of ground military vehicles, i.e., armored personnel carrier: BMP-2, BRDM-2, BTR-60, and BTR-70; tank: T62 and T72; rocket launcher: 2S1; air defense unit: ZSU-234; truck: ZIL-131; and bulldozer: D7. Fig. 9 shows the optical image and SAR image of each target.

The MSTAR data set includes targets at different depression angles. As a matter of routine, samples at the depression angle of 17° serve as the training set. For the few-shot SAR ATR experiments in this article, the data set needs to be divided into the training set, the support set, and the query set. In order to test the recognition performance of HIN under different depression angle settings, two groups of experiments are designed, as shown in Table I. For the sake of performance evaluation under different depression angle settings, samples at the depression angle of 17° serve as the training set and the support set, while samples at the depression angles of 15° and 30° serve as the query set, respectively.

For a typical experimental setting in the MSTAR data set [38], the existing algorithms are tested by three categories of targets, i.e., 2S1, BRDM-2, and ZSU-234. For the convenience of comparison, we follow this setting and select these three categories of targets as the support set and the query set. For the query set, the categories of the depression angles 15° and 30° contain 274 and 287 samples, respectively. Specifically, 1 or 5 (as a matter of routine, 1-shot and 5-shot are common few-shot experimental settings [25]–[28].) samples of each category are randomly selected as the support set in each experiment.

Then, the remaining seven categories of targets at the depression angle of 17° are selected as the training set. The training set consists of BMP-2, BTR-70, T-72, BTR-60, D-7, T-62, and ZIL-131. Because the categories of the targets differ in the sample number for the MSTAR data set, to ensure the balance of the samples, each category of the training set contains 200 samples.

Detailed information of the training set, the support set, and the query set is listed in Tables II and III. It should be

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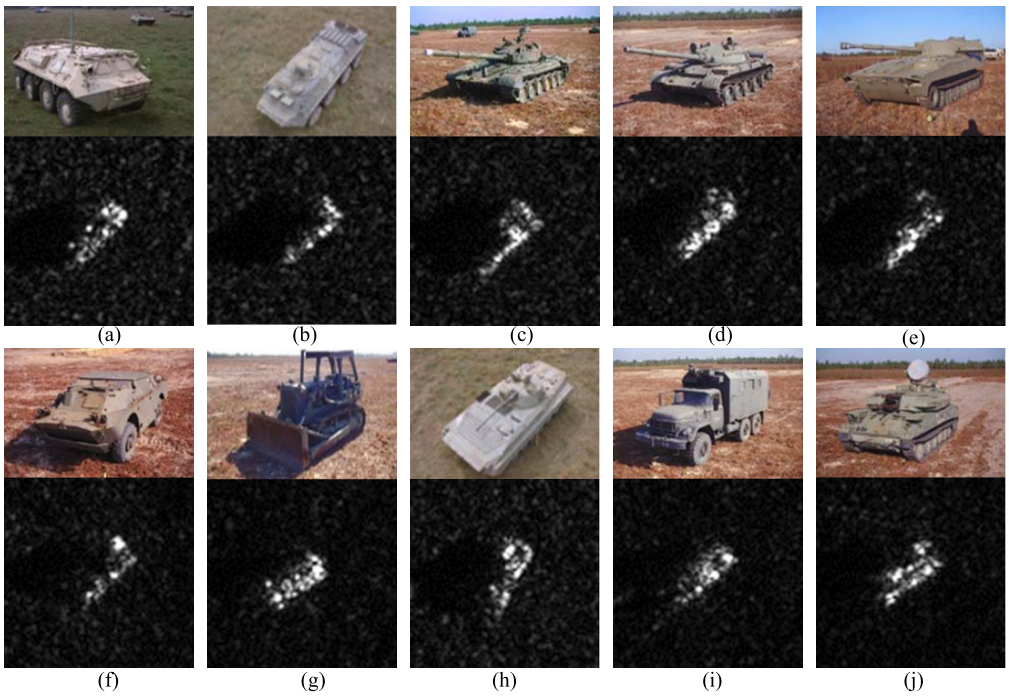


Fig. 9. Ten types of targets in MSTAR data set. (Top) Optical images versus (Bottom) SAR images. (a) BTR-60. (b) BTR-70. (c) T-72. (d) T-62. (e) 2S1. (f) BRDM-2. (g) D-7. (h) BMP-2. (i) ZIL-131. (j) ZSU-234.

TABLE II
TRAINING SET FOR THE EXPERIMENTAL SETUP

	Training set	
Class	Depression	Sample No.
BMP-2	17°	200
BTR-70	17°	200
T-72	17°	200
BTR-60	17°	200
D-7	17°	200
T-62	17°	200
ZIL-131	17°	200

TABLE III
SUPPORT SET AND QUERY SET FOR THE EXPERIMENTAL SETUP

Experiment		Support set		Query set	
	Class	Depression	No.	Depression	No.
1	2S1	17°	1/5	15°	274
	BRDM-2	17°	1/5	15°	274
	ZSU-234	17°	1/5	15°	274
2	2S1	17°	1/5	30°	287
	BRDM-2	17°	1/5	30°	287
	ZSU-234	17°	1/5	30°	287

noted that although the model parameters, the training set, and the query set remain constant, different support sets lead to different recognition accuracies. Therefore, each experiment is repeated 1000 times for randomly selected support sets.

In addition, Zhou *et al.* [39] and Schumacher and Schiller [40] showed that the clutter in the training and the test images of the MSTAR data set is highly correlated. Thus, experiments are carried out on the clutter-removed images, where the impact of similar background clutter on the classification is removed [39]. Moreover, the 128×128 image is cropped to 60×60 to reduce the computational burden since the target region only occupies a small region at the center of the original image.

B. Recognition Results of HIN

For the support and the query sets shown in Table III, the recognition results of the proposed method are shown in Table IV. In Table IV, the minimum, the maximum, the mean, and the standard deviation of the recognition accuracy for each group of the 1000 experiments are listed. For 1-shot SAR ATR tasks, the minimum accuracies of each group of experiments are only 35.65% and 42.86% and much lower than their mean accuracies. It indicates that the unrepresentative support set will decrease the recognition performance of HIN. In 5-shot FSL tasks, the robustness of HIN is greatly improved and the minimum accuracies of each group of experiments are increased to 83.70% and 78.63%. It indicates that although there are some unrepresentative samples, the rest samples can still be utilized for classification effectively in 5-shot SAR ATR tasks. The maximum accuracy reveals another aspect of how the support set affects HIN. For 1-shot SAR ATR tasks, the maximum accuracies of each group of experiments are 98.66% and 92.33%. As long as the samples in the support set are representative enough, even there is only one support sample for each class, the recognition performance of HIN is

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TABLE IV

STATISTICAL RESULTS OF RECOGNITION RATE OF HIN

Experiment	Inference Method		Accuracy			Standard Deviation
			min	max	mean	
1	Hybrid Inference	1shot	35.65	98.66	88.85	0.0793
		5shot	83.70	99.76	96.67	0.0249
2	Hybrid Inference	1shot	42.86	92.33	82.40	0.0676
		5shot	78.63	97.91	90.08	0.0241

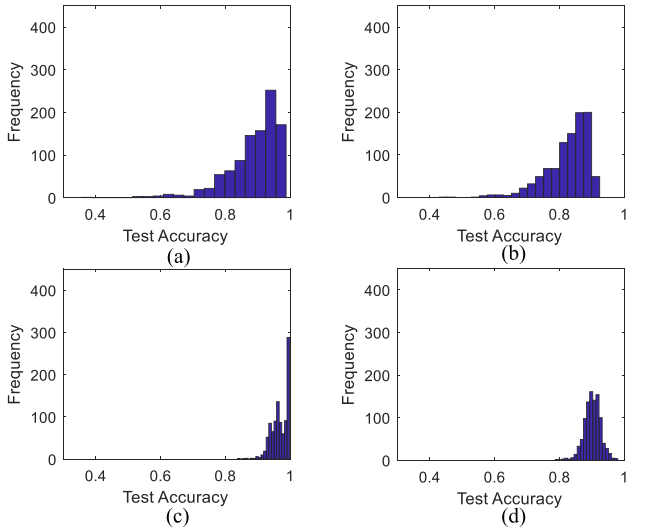


Fig. 10. Statistical histogram of the recognition accuracy for each group of 1000 independent experiments. (a) Experiment 1 1shot. (b) Experiment 2 1shot. (c) Experiment 1 5shot. (d) Experiment 2 5shot.

TABLE V

CONFUSION MATRIX OF THE AVERAGE RECOGNITION ACCURACY

	Test on 15°			Test on 30°		
1shot	2S1	BRDM2	ZSU234	2S1	BRDM2	ZSU234
2S1	72.37	25.53	2.10	58.95	34.66	6.38
BRDM2	2.39	97.61	0	5.60	94.13	0.28
ZSU234	3.39	0.04	96.57	5.84	0.03	94.13
5shot	2S1	BRDM2	ZSU234	2S1	BRDM2	ZSU234
2S1	91.18	6.39	2.43	72.26	18.87	8.86
BRDM2	0.76	99.24	0	1.64	98.01	0.35
ZSU234	0.40	0.01	99.59	0.05	0	99.95

still satisfying. To further analyze the effect of the number of samples in the support set on the recognition, the histogram of the recognition accuracy for each group of 1000 independent experiments is plotted in Fig. 10. In each group of experiments, the accuracy distribution of 5-shot tasks is more concentrated than that of 1-shot tasks. Accordingly, the standard deviation of 5-shot tasks is smaller than that of 1-shot tasks, as shown in Table IV.

TABLE VI

RECOGNITION RESULTS OF INDUCTIVE INFERENCE AND TRANSDUCTIVE INFERENCE

Experiment	Inference Method		Accuracy			Standard deviation
			min	max	mean	
1	Inductive Inference (Euclidean Distance)	1shot	36.25	96.35	83.29	0.1012
		5shot	65.33	98.66	91.71	0.0564
	Transductive Inference (Label Propagation)	1shot	15.57	99.51	88.36	0.0853
		5shot	79.32	99.64	96.53	0.0263
2	Inductive Inference (Euclidean Distance)	1shot	39.72	84.67	70.87	0.0961
		5shot	54.24	92.92	82.11	0.0774
	Transductive Inference (Label Propagation)	1shot	23.35	99.42	79.18	0.0853
		5shot	72.13	99.88	90.28	0.0568

It is well known that SAR images are extremely sensitive to the variance of depression angles. The comparison between experiment 1 and experiment 2 indicates that depression angle is an important factor that affects the recognition performance of HIN. Because the maximum and the minimum have great contingency, we focus on the average recognition accuracy in the experimental analysis, and the confusion matrices of the average recognition accuracy of each group of experiments are shown in Table V. The average recognition accuracy of experiment 1 is higher than that of experiment 2 in both 1-shot and 5-shot tasks. Therefore, smaller variance of depression angles between the query set and the support set is beneficial for improving the recognition accuracy. Table V also suggests that the 2S1 target is often misclassified to the BRDM-2 target with high probabilities. In our opinion, it is caused by the similarity of different types of targets at different azimuth angles. Specifically, SAR images of the same type vary greatly at different azimuth angles and the scattering characteristics between these two targets may be quite similar sometimes.

C. Effectiveness of the Hybrid Inference

In this section, the effectiveness of hybrid inference is further illustrated by comparing its recognition performance to that of inductive inference and transductive inference. The recognition results of inductive inference and transductive are shown in Table VI. The inductive inference method only utilizes the Euclidean distance-based classifier and the classification result of each query sample is output by (2). L_E defined by (3) is utilized as the loss function. Similarly, the transductive inference method only utilizes label propagation for classification and the classification result of each query sample is output by (9). L_{lp} defined by (10) is utilized as the loss function. In Table VI, the minimum, the maximum, the mean, and the standard deviation of the recognition accuracy for each group of the 1000 experiments are listed. For the data set in Table III, the average recognition accuracy of inductive

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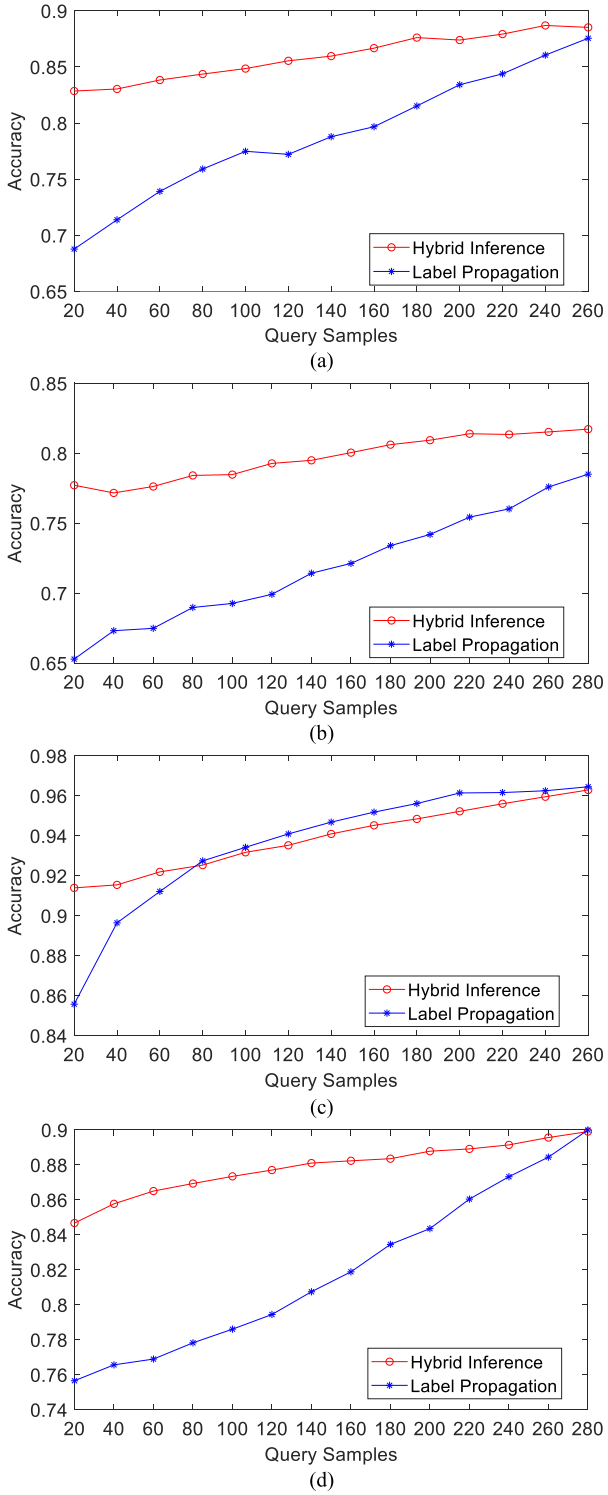


Fig. 11. Influence of the number of query samples on the classification performance of hybrid inference and label propagation. (a) Experiment 1 1shot. (b) Experiment 2 1shot. (c) Experiment 1 5shot. (d) Experiment 2 5shot.

inference is much lower than that of hybrid inference shown in Table IV. However, the average recognition accuracy of transductive inference is close to that of hybrid inference in Table IV because transductive inference plays a major role in hybrid inference when there are sufficient query samples.

To further study the influence of the number of query samples on transductive inference and hybrid inference, different number of query samples are randomly sampled from the query set in Table III as new query sets and each experiment is repeated 1000 times independently. Fig. 11 shows the influence of the number of query samples on the classification performance of hybrid inference and label propagation. The coordinates on the horizontal axis represent the number of query samples for each type of target. The accuracy in Fig. 11 is the mean accuracy of 1000 independent experiments. Because the recognition accuracy of inductive inference has nothing to do with the number of query samples, the effect of the number of query samples on inductive inference is not discussed. As shown in Fig. 11, the increase of the number of query samples is beneficial to improve the recognition accuracy of both hybrid inference and label propagation. It is easy to understand that sufficient query samples are beneficial for label propagation to make better use of the manifold structure of the query set; thus, the classification accuracy is improved. Since label propagation is an important part of hybrid inference, more query samples also improve the classification accuracy of hybrid inference. In addition, it is observed that the number of query samples has a much greater impact on label propagation than hybrid inference. The recognition performance of hybrid inference can be effectively improved with the help of Euclidean distance-based classifier when the number of query samples is small. As discussed above, hybrid inference takes the advantage of both inductive inference and transductive inference while avoiding their shortcomings. When there are sufficient query samples, the recognition performance of hybrid inference is improved by transductive inference. When there are insufficient query samples, inductive inference alleviates the degradation of recognition performance.

TABLE VII
COMPARISON BETWEEN HIN AND OTHER SAR ATR METHODS

Method	no. training /support	15° test acc (%)	30° test acc (%)
SRC	896	98.18	96.14
KSRC	896	99.03	97.22
KLR	896	98.66	96.86
MSRC	896	99.64	98.29
SUM	896	99.88	96.58
MAP	896	99.88	97.22
CKLR1	896	99.63	97.12
CKLR2	896	99.75	98.29
HIN(1shot)	3	**98.66**	**92.33**
HIN(5shot)	15	**99.76**	**97.91**

D. Comparison Between HIN and Traditional SAR ATR Methods

In order to further analyze the performance of the proposed algorithm, the comparison between HIN and traditional SAR ATR methods is given in Table VII. For the three-class classification problem on the MSTAR data set, Dong and Kuang [38] give the classification results of eight methods, i.e., sparse representation-based classifier (SRC) [41], kernel sparse representation-based classifier (KSRC) [42], kernel linear representation-based classifier (KLR) [43], sparse representation of monogenic (MSRC) [44], score level fusion via

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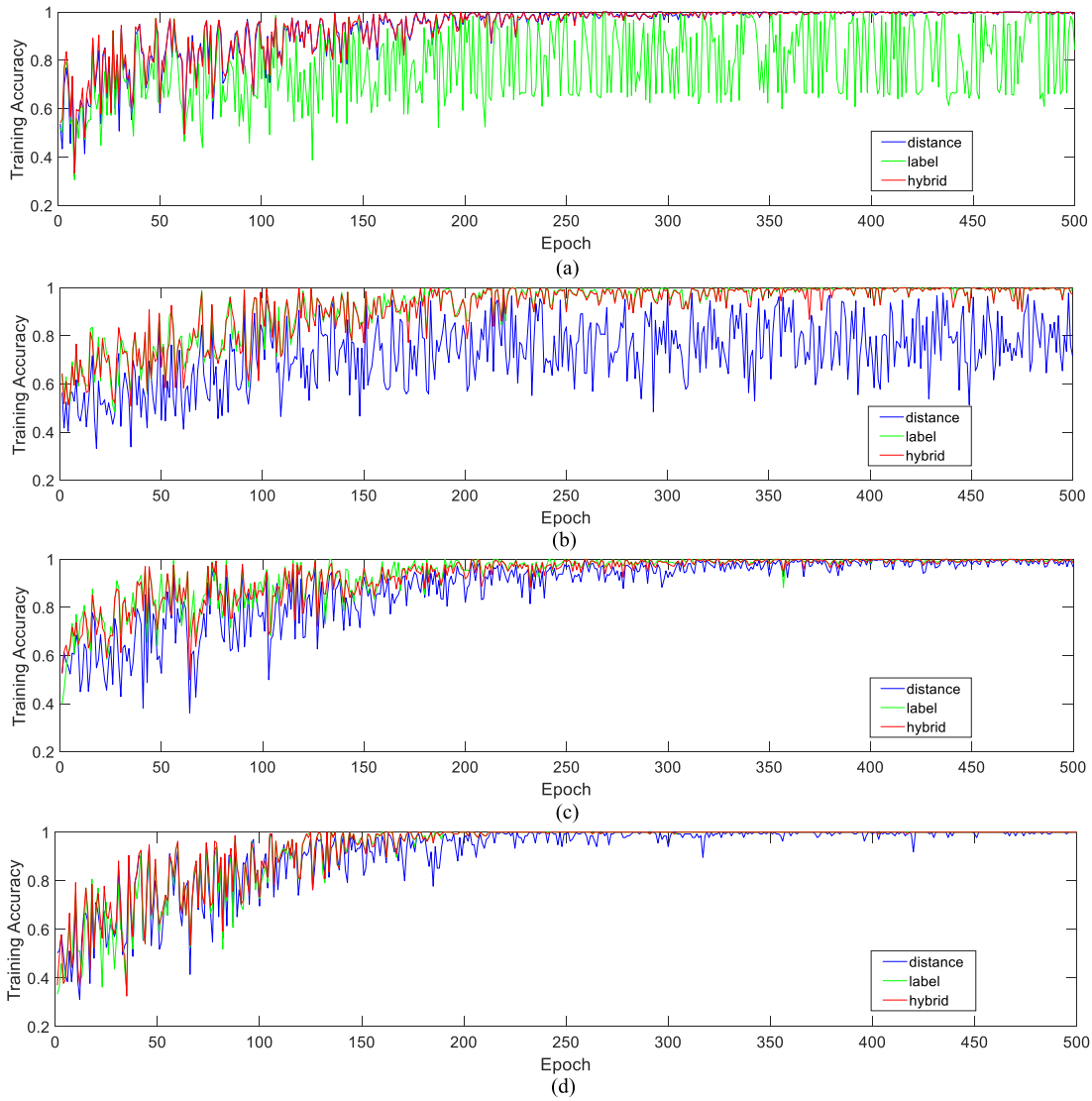


Fig. 12. Training accuracy of three inference methods with different loss functions for training. (a) Train by L_E. (b) Train by L_{lp}. (c) Train by L_h. (d) Train by L_{eh}.

summation rule (SUM), score level fusion via MAP criteria (MAP), kernel combination via stacked features (CKLR1), and kernel combination via summation kernel (CKLR2).

These methods select 2S1, BRDM2, and ZSU234 from the MSTAR data set at the depression angle of 17° as the training set, with each category containing 299, 298, and 299 training samples, respectively.

Since different support sets will lead to different recognition results, the best result of the 1000 random tests is chosen for illustration. Because the training set in [38] contains the support set in HIN, and has a large number of samples, such comparison is fair. As shown in Table VII, the performance of HIN is comparable to these traditional SAR ATR methods for the three-class classification problem although only 1 or 5 samples of each class are adopted as the support set, demonstrating the robustness to the extreme scarcity of samples.

E. Effectiveness of the Enhanced Hybrid Loss

In this section, the effectiveness of the enhanced hybrid loss will be illustrated. Fig. 12 shows the variation of the training accuracy in 500 training epochs with different loss functions. The blue line, the green line, and the red line represent the classification accuracies of the Euclidean distance-based classifier, label propagation, and hybrid inference, respectively. As shown in Fig. 12(a), if the network is trained by L_E, the Euclidean distance-based classifier will have a high classification accuracy, while the classification accuracy of label propagation will be much lower. It indicates that L_E cannot constrain the embedding network to learn an embedding space where the samples of the same class get close to one another. Analogously, Fig. 12(b) indicates that L_{lp} cannot fully utilize the labeled samples in the support set and the classification accuracy of the Euclidean distance-based classifier is not satisfying. Fig. 12(c) shows that L_h gets a balance between the two classifiers and none of the classifiers outputs poor classification accuracy. Further, Fig. 12(d) shows that L_{eh} strengthens the role of each inference method and effectively improves the stability of convergence in the training procedure.

In order to explain the effectiveness of L_{eh} more clearly, a common high-dimensional data visualization method, i.e.,

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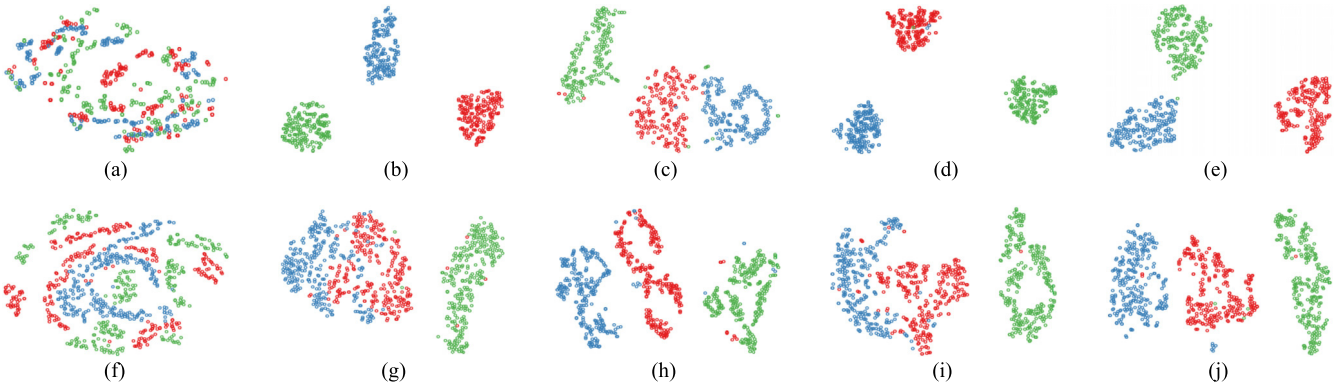


Fig. 13. Visualization of 2S1, BRDM-2, and ZSU-234 by t-SNE. (a) Initial support set. (b) Support set in the embedding space trained by L_E. (c) Support set in the embedding space trained by L_{lp}. (d) Support set in the embedding space trained by L_h. (e) Support set in the embedding space trained by L_{eh}. (f) Initial query set. (g) Query set in the embedding space trained by L_E. (h) Query set in the embedding space trained by L_{lp}. (i) Query set in the embedding space trained by L_h. (j) Query set in the embedding space trained by L_{eh}.

t-SNE [45], is utilized to visualize the original SAR images and the embedding vectors of HIN trained by different loss functions. Fig. 13(a) and (f) shows samples of the support set and the query set in the original SAR image space. As shown in Fig. 13(b) and (g), in the embedding space trained by L_E, the query set has much poorer separability than that of the support set. It indicates that L_E causes the poor generalization performance of HIN. As shown in Fig. 13(c) and (h), the support set in the embedding space trained by L_{lp} does not have good separability and the embedding vectors of the query set have large intraclass variance. Fig. 13(d) and (i) shows that although HIN trained by L_h performs well on the support set, the interclass variance of the query set in the embedding space is small and effective classification becomes difficult. To our relief, Fig. 13(e) and (j) shows that the samples of the support set and the query set both have small intraclass variation and large interclass separation in the embedding space trained by L_{eh}.

V. Conclusion

A novel FSL method, namely, HIN, is proposed in this article, which overcomes the deficiency of singly adopting inductive inference or transductive inference in FSL. By taking a hybrid inference method, HIN can effectively utilize the supervised information of the labeled samples and the manifold structures of the unlabeled samples. In addition, an enhanced hybrid loss that constrains samples to have small intraclass variation and large interclass separation in the embedding space is proposed to train HIN. Experimental results on the MSTAR data set have illustrated the effectiveness and the robustness of HIN in few-shot SAR ATR tasks.

Future work will be focused on zero-shot SAR ATR algorithms, which could recognize new target types not included in the support set.

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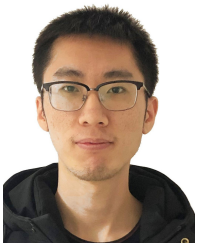
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Li Wang was born in Jiangsu, China, in 1992. He received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2015 and 2020, respectively.

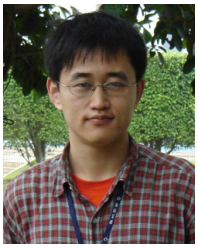
He is a Lecturer with the Ministry Key Laboratory of Electronic Information Countermeasure and Simulation, Xidian University. His major research interests include deep learning and radar automatic target recognition.



Xueru Bai (Member, IEEE) was born in Xi'an, Shaanxi, China, in 1984. She received the B.S. and Ph.D. degrees in signal and information processing from Xidian University, Xi'an, in 2006 and 2011, respectively.

She is a Professor with the National Laboratory of Radar Signal Processing, Xidian University. Her research interests include high-resolution radar imaging and radar automatic target recognition.

Dr. Bai was a recipient of the National Excellent Doctoral Dissertation Award granted by the Ministry of Education of China and the Program for Excellent Young Scientist selected by the National Natural Science Foundation of China.



Chen Gong (Senior Member, IEEE) received the B.S. degree in electrical engineering and mathematics (minor) from Shanghai Jiao Tong University, Shanghai, China, in 2005, the M.S. degree in electrical engineering from Tsinghua University, Beijing, China, in 2008, and the Ph.D. degree in electrical engineering from Columbia University, New York City, NY, USA, in 2012.

He was a Senior Systems Engineer with Qualcomm Research, San Diego, CA, USA, from 2012 to 2013. He is a Faculty Member with the University of Science and Technology of China, Hefei, China. His research interests include wireless communications, optical wireless communications, and signal processing.



Feng Zhou (Member, IEEE) was born in Tongxu, Henan, China, in 1980. He received the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2007.

He is the Director and a Professor with the Key Laboratory of Electronic Information Countermeasure and Simulation Technology of Ministry of Education, Xidian University. He has authored or coauthored over 80 articles. His research interests include high-resolution radar imaging and radar countermeasures.

Dr. Zhou was a recipient of the Young Scientist Award from the XXXI URSI GASS Committee, the Program for Support of Top-Notch Young Professionals from the Central Organization Department of China, and the Program for New Century Excellent Talents in University from the Ministry of Education of China.

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